(Please write your Enrolment Number immediately) Enrolment No._____

MID TERM Examination

4th Semester, Mar 2025

Paper Code: CIC-206
Time: 1.5 Hrs.

Subject: Theory of Computation
Max Marks: 30

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining

Q1. Attempt any FIVE of the following. (2*5)

a) What is the difference between Deterministic Finite Automata (DFA) and Non-Deterministic Finite Automata (NDFA)? [CO1]

b) Define the term context free grammar with an example. [CO2]

c) Define instantaneous description of PDA. [CO2]

d) What are the disadvantages of unambiguous parse tree? Give an example. [CO2]

e) Describe the set of all Strings of 0's and 1's ending in 00 by Regular Expression. [CO1]

f) Write any five identity rules for regular expressions. [CO1]

Q2. a) Construct the DFA equivalent to NDFA whose transition table is given below (q0 – initial state, q4 – final state): (5) [CO1] Present

State	0	1	2
q0		q1, q4	q2	q2, q3
q1		-	q4	-
q2		q1, q3	-	q2, q4
q3		-	q4	-
q4		-	q1, q3	-

b) Prove the pumping lemma for regular languages with example. (5) [CO1]

Q3. a) Design a PDA for the language $\{a^n b^n, n>1\}$. (5) [CO2]

b)Construct a equivalent grammar G in CNF for the grammar G1 where G1=({S,A,B},{a,b},{S->bA/aB,A->bAA/aS/a,B->aBB/bS/b},S). (5) [CO2]

Q4. a) Explain Chomsky's Hierarchy of grammar in detail. (5 Marks) [CO1]

b) Prove that deterministic and non-deterministic PDA's are not equivalent.(5) [CO2]